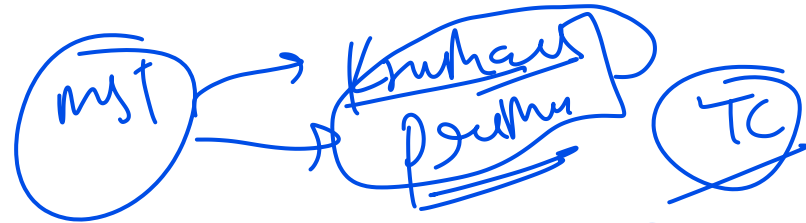


2-3

Spanne Tree



Lec_68

GATE Questions on MST

PMS

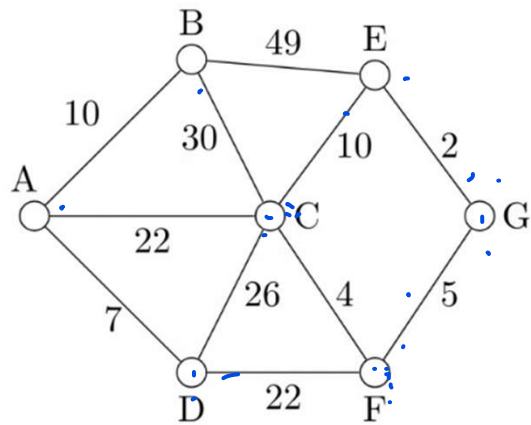
Numerical

think

mst
30
1 can
1 quib

1.GATE 2004

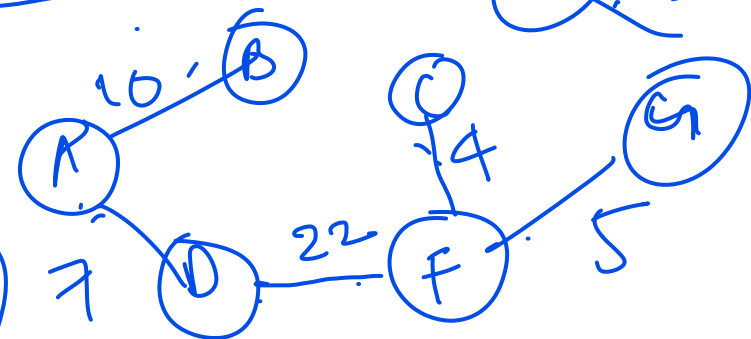
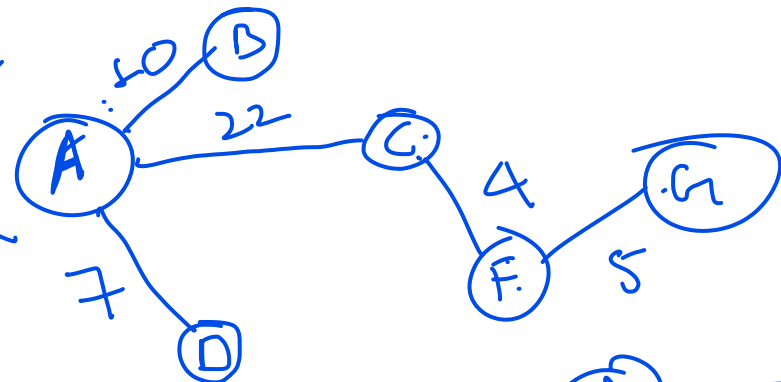
Consider the undirected graph below:



Using Prim's algorithm to construct a minimum spanning tree starting with node A, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

- ☒ A. (E, G), (C, F), (F, G), (A, D), (A, B), (A, C)
- ☒ B. (A, D), (A, B), (A, C), (C, F), (G, E), (F, G)
- ☒ C. (A, B), (A, D), (D, F), (F, G), (G, E), (F, C)
- ☒ D. (A, D), (A, B), (D, F), (F, C), (F, G), (G, E)

$V=7$
 $E=6$



Knicks

and Penny

Working

Numerical

understood

msI

2.GATE 2024

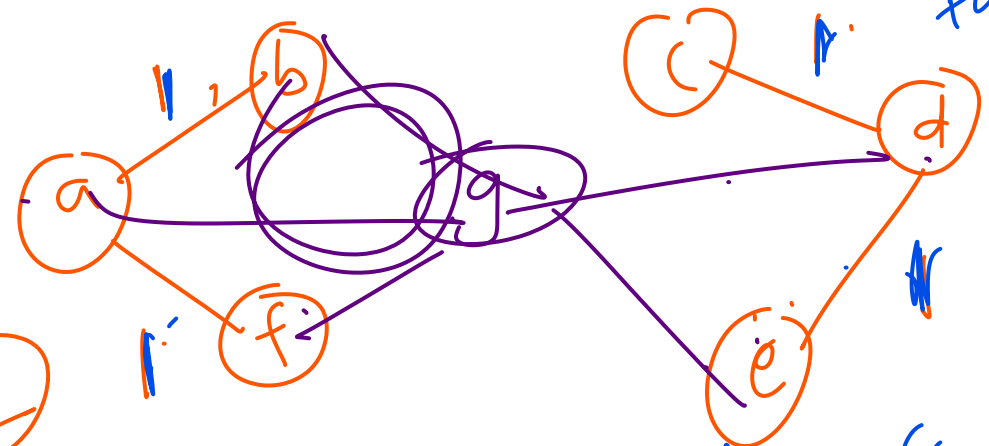
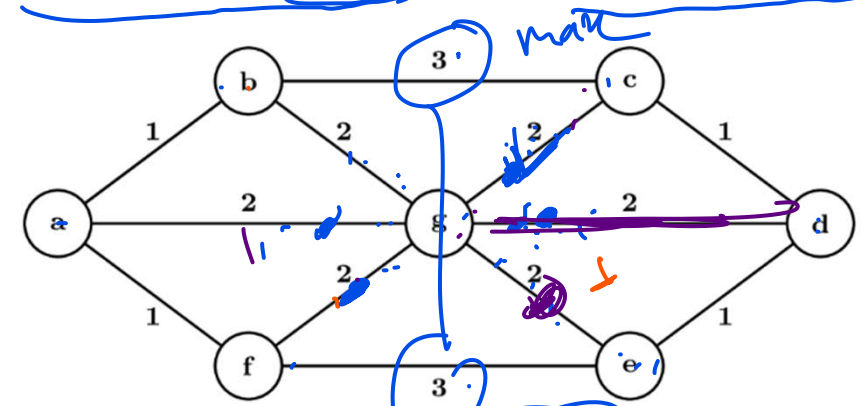
mst

Set 2

⇒ 30 second

mst ⇒ minimize the weight of the tree

The number of distinct minimum-weight spanning trees of the following graph is



9

m1

we have

6 edges

$$6C2 = \frac{6!}{2!4!} = \frac{6 \times 5}{2} = 15$$

$$15 - 6 = 9$$

$$3 + 3 + 3$$

$$V=7$$

$$E=6$$

⇒ mst

2

9

9

$\Rightarrow$

$m1$

$\Rightarrow$

loop

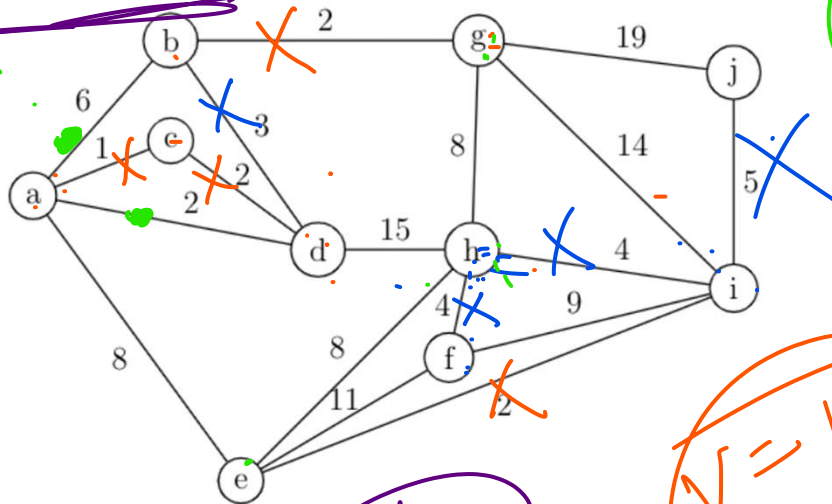
$m2$

$\Rightarrow$

~~branch~~

3.Gate 2003

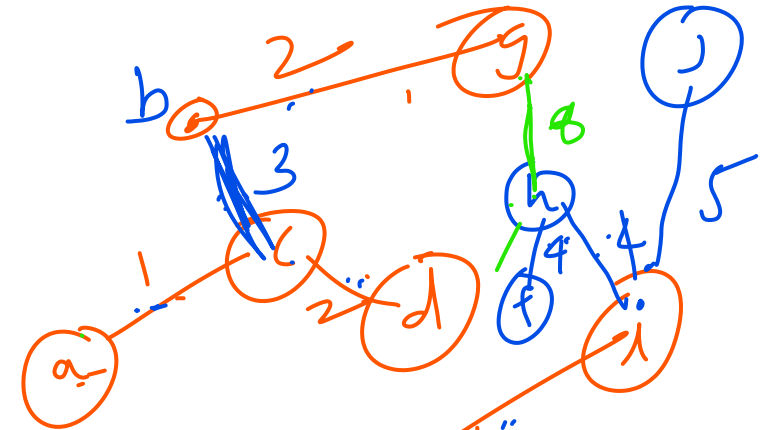
What is the weight of a minimum spanning tree of the following graph?



min. tree

Kruskal

2-3



$V = 10$
 $E = 9$

Cost 23

Kruskal's algorithm

Cancel

4.) GATE 2021 Set2

Let G be a connected undirected weighted graph. Consider the following two statements.

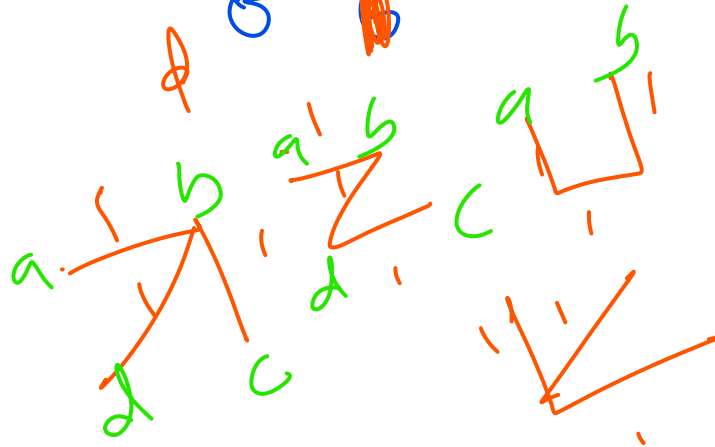
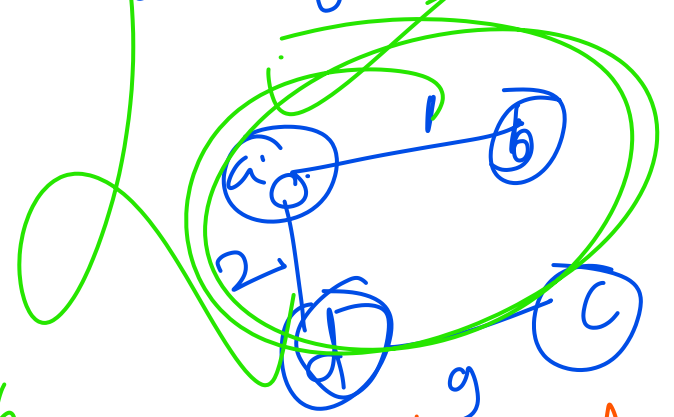
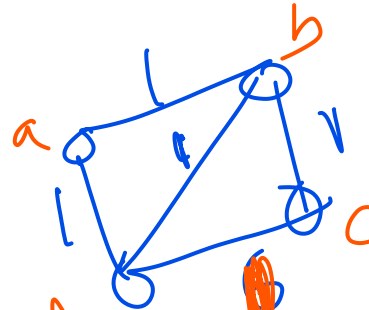
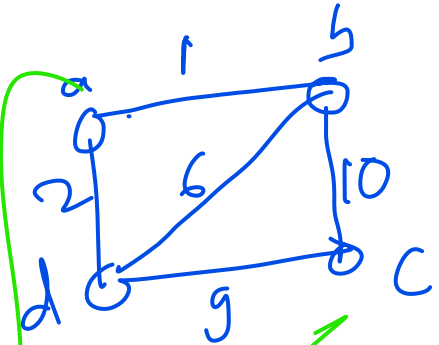
S_1 : There exists a minimum weight edge in G which is present in every minimum spanning tree of G .

S_2 : If every edge in G has distinct weight, then G has a unique minimum spanning tree.

Which one of the following options is correct?

- A. Both S_1 and S_2 are true
- B. S_1 is true and S_2 is false
- C. S_1 is false and S_2 is true
- D. Both S_1 and S_2 are false

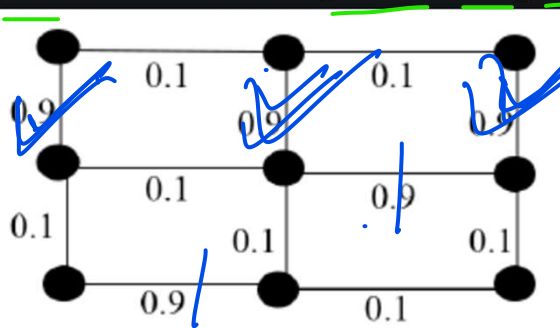
iff all edges have same



5.) GATE 2021 set1

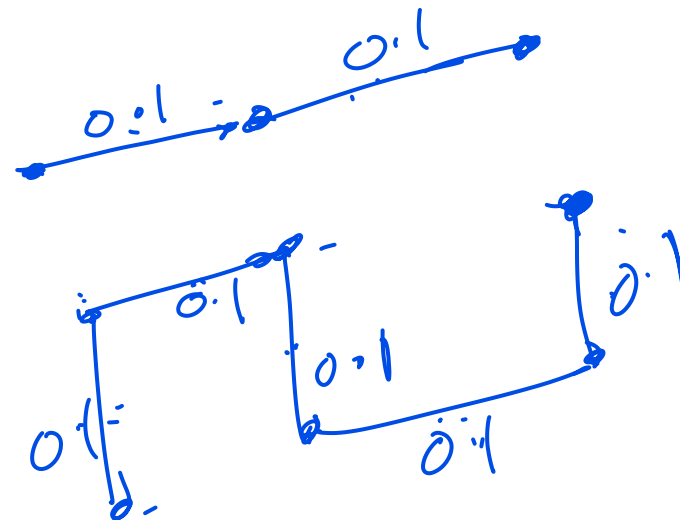
$E = 17$

Consider the following undirected graph with edge weights as shown:

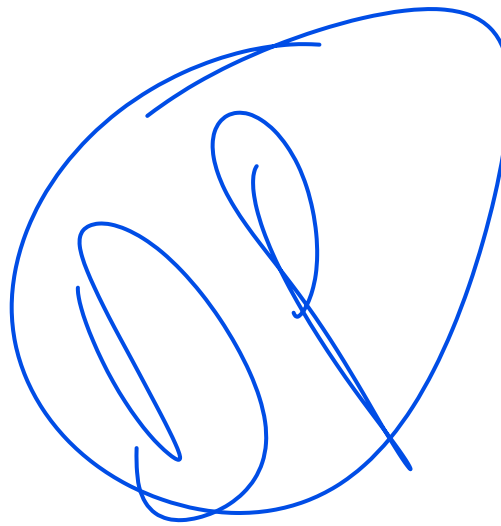


The number of minimum-weight spanning trees of the graph is _____.

3



$V = 9$
 $E = 17$



gate 18

filter
nella fu

